

Too Far to Matter? The Urban Reach of Rural Territorial Conflict

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Can persistent territorial conflict affect nearby urban economies even when conflict events occur outside the city? I study this question in southern Chile, where the Mapuche self-determination conflict is recurrent and predominantly rural. I link geocoded attacks, coercive actions, and land invasions to a panel of 104 fixed urban areas and measure annual external exposure from events occurring within 25 kilometers but outside the receiving area. Greater external conflict exposure is associated with fewer registered enterprises and lower tax-reported firm sales. Within an urban area, an increase from the 25th to the 75th percentile of positive annual exposure is associated with about 2.2 percent fewer enterprises and 5.3 percent lower sales. First registrations also decline with exposure, pointing to weaker formal entry as an adjustment margin. Recorded terminations do not indicate greater exit. Complementary measures of nighttime lights and newly built area show no comparable contemporaneous decline. Fiscal appraisal also does not decline contemporaneously, but turns negative at medium lags. The formal-activity association is not systematically stronger where historical Títulos de Merced—the state titles that demarcated Mapuche reservations and now locate historically recognized territorial claims—cover more surrounding land, or where the baseline Mapuche population share is higher pointing to an association of the conflict itself and not to the Indigenous presence. The results show that the economic reach of recurrent territorial conflict can extend beyond contested land and into nearby urban economies, even in a setting where property, tax, Indigenous-land protection, and restitution institutions continue to operate.

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JEL codes: D74; O18; R11.

1 Introduction

Conflict is usually studied where violence takes place, but its economic consequences need not stop at the location of the event. Firms in nearby towns may respond when recurrent conflict changes perceived security, access to customers and suppliers, or expectations about the future use of surrounding land. This paper asks whether persistent territorial conflict reaches nearby urban economies even when the conflict itself occurs outside them.

A large literature documents the economic consequences of conflict for directly exposed places and firms. Armed conflict reduces firm sales and production inputs, increases plant exit, and can alter employment and investment, while violence and insecurity are capitalized into property and other assets (Camacho and Rodriguez, 2013; Del Prete et al., 2023; Besley and Mueller, 2012; Brancati et al., 2026). Related evidence from Afghanistan and Mexico shows that the response of firms also depends on how violence affects security, state capacity, and local labor markets (Neri-Lainé, 2026; Utar, 2025). Much of this evidence, however, concerns firms located where conflict or insecurity occurs. A smaller literature studies how violence propagates across space. Alfano and Cornelissen (2026) show that attacks along transport routes in Somalia affect food prices in distant destination markets, while Liu et al. (2026) emphasize how trade-network connectivity conditions the transmission of violence across Nigerian markets. These studies show that the economic geography of conflict can be broader than the geography of violence itself, but they observe specific trade connections through which the shock propagates. Less is known about whether recurrent conflict around a territory affects nearby urban economic activity.

Southern Chile provides a distinct setting in which to study this question. The Mapuche conflict is an Indigenous self-determination conflict centered on territorial claims, land restitution, and political recognition rather than a conventional civil war (Cayul et al., 2022). Its geography reflects the military incorporation of Mapuche territory into the Chilean state in the late nineteenth century, the creation and subsequent fragmentation of *Títulos de Merced*, the reversal of twentieth-century land transfers, and the persistence of claims over land later transferred to non-Mapuche owners (Jaimovich and Jordán, 2025b,a; Jaimovich and Toledo-Concha, 2026). Contemporary attacks and land invasions are therefore rooted in a historically structured geography of territorial claims. Of the 2,775 eligible conflict events, 2,288 (82.5 percent) occur outside all study urban areas.

The institutional environment is also unusual relative to many settings in the conflict literature. Contemporary territorial conflict does not take place in an institutional vacuum. The 1993 Indigenous Law regulates Indigenous land transfers and created a public registry through which Indigenous-land status is enforced; CONADI operates a formal land fund through which the state purchases and restores land associated with recognized claims (Gobierno de Chile, 1993; Jordán et al., 2025; Jaimovich et al., 2025). Registry protection has slowed the loss of Indigenous ownership, while restitution has changed some land uses and continued during the conflict (Jordán et al., 2025; Jaimovich et al., 2025). Related work from other Indigenous settings shows why this distinction matters. Legal recognition and property institutions can alter bargaining power and economic outcomes (Feir et al., 2023). Albertus (2026) shows that recognition during Peru’s civil conflict reduced violence partly by moving land disputes into state institutions. Southern Chile therefore makes it possible to study whether recurrent territorial conflict has economic costs beyond contested land even when the state continues to register property, collect taxes, regulate Indigenous land, and operate a formal restitution process.

This paper also differs from the existing evidence on the economic cost of the Mapuche conflict. Larroulet Vignau and Sepúlveda (2025) estimate aggregate losses in regional GDP associated with

violence in Araucanía. I instead exploit repeated variation in the location and intensity of conflict across 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. This four-region corridor is selected because historical Mapuche and Huilliche territorial occupation, recognized claims, contemporary restitution, and recent conflict overlap there (Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas, 2008; Jaimovich and Jordán, 2025b; Jaimovich et al., 2025; Cayul et al., 2022). Appendix Figure A1 shows the corridor alongside a historical map of Mapuche territory and the 1992 Mapuche population share. Each area is constructed from the 2017 INE urban census-block geography and held fixed over time; these are geographic areas, not functional labor-market areas. The unit of interest is the receiving urban area rather than the region where the conflict occurs. This makes it possible to separate conflict taking place inside an urban area from conflict occurring nearby, and to ask directly how far the association with economic activity extends. I also compare formal business outcomes with nighttime lights, fiscal appraisal, and newly built area. This distinction helps establish whether the economic reach of nearby conflict appears mainly in administrative records of formal activity or is accompanied by broader changes in the physical development of cities.

I link these fixed urban areas to geocoded events from the Mapuche Conflict Event and Data Archive (MACEDA), focusing on attacks, coercive actions, and land invasions (Cayul et al., 2022). External exposure counts events occurring outside the receiving area at an exact minimum straight-line distance of at most 25 kilometers from its footprint; conflict inside the area enters separately. The 25-kilometer measure is intended to map nearby external exposure, not to identify the specific channel through which one location affects another. Geographic proximity can proxy for several forms of economic connectivity and does not by itself identify a transmission mechanism (Debarys and Le Gallo, 2025). The baseline estimates compare changes within the same urban area over time, absorb region-specific annual shocks, condition on contemporaneous within-area conflict, and allow trajectories to differ with pre-sample population, rurality, and historical conflict exposure.

Greater external conflict exposure is associated with weaker formal business activity. An increase from the 25th to the 75th percentile of positive annual exposure is associated with 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported firm sales. The enterprise association also appears in the timing evidence: coefficients remain negative over the first several years after exposure, while future exposure does not systematically predict current enterprise outcomes. Sales dynamics are less precisely estimated. Fiscal appraisal, nighttime lights, and newly built area do not show a corresponding contemporaneous contraction. Fiscal appraisal turns negative at medium lags, consistent with its interpretation as a slowly updated assessed stock rather than a high-frequency market price. First registrations also decline, while recorded terminations provide no evidence of increased exit; an administrative break in the termination series limits comparisons over time. Within-area grid comparisons show negative associations between exposure and selected legal-entity counts at 5 and 10 kilometers and commercial-use share at 10, 20, 25, and 30 kilometers; nighttime lights, appraisal, and newly built area show no common negative gradient (Appendix D).

Historical territorial geography provides an additional distinction. The formal-business asso-

ciation does not become systematically stronger where *Títulos de Merced* cover more of the surrounding territory or where the 1992 Mapuche population share is higher. These results do not imply that historical claims or Indigenous presence are economically irrelevant. Rather, they provide no evidence that the negative association with contemporary conflict exposure is mechanically concentrated in places with greater historical Mapuche presence. The data also do not identify a single transmission channel. Security concerns, access to customers and suppliers, expectations about territorial disputes, and other forms of local connectivity remain plausible mechanisms.

The paper makes three contributions. First, it documents a spatial margin of territorial conflict that is difficult to observe from either conflict locations or regional aggregates alone: recurrent conflict outside an urban area is associated with formal economic activity inside nearby cities. Second, it adds a distinct conflict setting to a literature dominated by armed conflict, civil war, terrorism, and severe criminal violence. The Mapuche case is a persistent Indigenous self-determination conflict that operates alongside an active institutional architecture of property registration, Indigenous-land protection, and restitution. Third, by combining formal business records with directly spatial urban outcomes and historical land institutions, the paper shows that the formal-activity association is not accompanied by a common contemporaneous physical decline and is not systematically stronger where historical claims or baseline Mapuche presence are greater. More broadly, the results suggest that the economic incidence of territorial conflict may extend beyond the land under dispute, so assessments of its local consequences should also consider nearby urban economies.

2 Institutional Setting and Conflict Geography

2.1 Territorial self-determination and historical claims

The Mapuche are an Indigenous group of south-central Chile whose historical territory encompasses much of the study corridor. Mapuche communities retained substantial autonomy through most of the colonial period and were incorporated into the Chilean state only after the military occupation of Araucanía in the late nineteenth century. Between 1884 and 1929, the state confined Mapuche communities to nearly 3,000 reservations covering roughly 550,000 hectares, a small share of their ancestral territory (Jaimovich and Jordán, 2025b; Jaimovich and Toledo-Concha, 2026). The state formalized these reservations through *Títulos de Merced* (TDM). The TDM therefore did not restore ancestral territory; they legally recognized the much smaller areas left to Mapuche communities after the military occupation. They created an important state-recognized geography of historical territorial claims that remains relevant to later land protection, restitution, and conflict. Importantly, TDM boundaries do not describe current Mapuche ownership. Through subsequent allotment, land sales, encroachment, weak enforcement, and later deregulated transfers, Mapuche communities lost ownership of substantial amounts of land within the historical reservations (Jaimovich and Jordán, 2025b,a). Contemporary claims can therefore concern land that lies within a historically recognized TDM but is no longer under Mapuche ownership. In the empirical analysis, TDM are used to locate this historical geography of recognized claims, not to identify the

current owner of a parcel or the ethnic identity of a nearby firm.

Agrarian reform extended the contested geography beyond the reservations. The reform began in 1962 and accelerated between 1970 and 1973, when Mapuche communities recovered or gained access to land through the expropriation of large estates. After the 1973 military coup, the dictatorship returned or reassigned much of that land, promoted forestry plantations, and divided remaining collective reservations into individual parcels. Contemporary invasions and attacks are more common on plots touched by that reform and its reversal, linking current conflict to grievances both within and beyond TDM boundaries (Jaimovich and Jordán, 2025a; Jaimovich and Toledo-Concha, 2026).

2.2 Formal land institutions and restitution

The return to democracy in 1990 introduced a new legal framework for Indigenous land claims. The 1993 Indigenous Law recognizes Indigenous land through legal history, including TDM, long-standing occupation, judicial recognition, and land later transferred by the state. It restricts alienation and encumbrance, although the rules differ between community and individually held land and permit specified transactions with authorization from the *Corporación Nacional de Desarrollo Indígena* (CONADI), the public agency created by the same law to implement Indigenous policy, including the protection and expansion of Indigenous land (Gobierno de Chile, 1993). The law also created the Public Registry of Indigenous Territories, which makes registration central to the enforcement of Indigenous-land status. Evidence from the Temuco property records shows that registered parcels experienced fewer post-inheritance sales and smaller losses of Mapuche ownership, while registry coverage declined over time (Jordán et al., 2025). The law also created an institutional mechanism for addressing historical land claims. CONADI administers the Fund for Indigenous Land and Water, including the mechanism established under Article 20(b), through which the state acquires land to address recognized Indigenous land claims. Between 1993 and 2024, CONADI acquired 230,563 hectares through this mechanism in Biobío, Araucanía, Los Ríos, and Los Lagos. The 2025 Presidential Commission for Peace and Understanding estimated that existing claims and applications corresponded to approximately 396,825 hectares under its proposed parameters, implying that the state had acquired an amount equivalent to about 58 percent of this institutionalized demand. Including estimated future demand increases this figure to approximately 486,000 hectares. These estimates refer to claims processed or potentially processed through the formal restitution system rather than to the full extent of historical Mapuche territory. Land restitution is therefore not only a historical response to dispossession but an ongoing state institution during the period of contemporary conflict.

2.3 Conflict geography and recurrence

The Mapuche Conflict Event and Data Archive (MACEDA) records geocoded events from local-media reports in the Mapuche self-determination conflict (Cayul et al., 2022). The main event

family contains attacks, coercive actions, and land invasions. Events recur over time rather than defining a single absorbing transition into conflict.

Most main events occur outside the study’s urban areas. Of the 2,775 main events in the four-region corridor, 2,288 (82.5%) lie outside all 104 fixed urban areas. La Araucanía contains 66.6 percent of these events despite accounting for 25.7 percent of the corridor’s 2017 population; conflict also occurs in Biobío, Los Ríos, and Los Lagos. Figure 1 maps the events and the locations of the receiving urban areas.

Annual event counts rise and fall throughout the period (Figure 2). Nearby urban areas may therefore face repeated changes in external exposure even when events occur beyond their own urban footprint.

3 Data & Empirical Strategy

3.1 Units, outcomes, and measurement

The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks (“manzanas” with $AREA=1$) within a municipality and remains fixed over time; its footprint can contain spatially disconnected components. These are geographic urban areas, not functional labor-market areas. Their median area is 4.59 km² (mean 8.21); the interquartile range is 2.45–7.60 km², and the full range is 0.61–63.91 km². The geographic support of the outcomes differs across data sources.

Annual SII records measure formal business activity associated with each urban area. Chile’s national tax authority (Servicio de Impuestos Internos, SII) reports enterprise stocks and tax-reported sales. Registered enterprises measure the stock of firms reporting activity in each municipality, while tax-reported sales capture the scale of formal market activity. Both series are available annually from 2005 to 2021. Because SII assigns firms to the municipality of their tax or headquarters address, these outcomes measure formal activity associated with each urban area, but they do not by themselves establish that the underlying activity occurs within the fixed urban footprint.¹

Nighttime luminosity provides a directly spatial measure of activity within the fixed urban footprints. Harmonized satellite imagery allows me to follow the same urban polygons over a longer period than the SII business records. Lights do not measure formal firms or sales directly, but they provide a complementary indicator of whether greater conflict exposure is accompanied by a broader change in urban economic activity.

The property cadastre captures a slower-moving margin. SII records allow me to measure both the total fiscal appraisal of properties within the fixed urban geography and mean appraisal per property. Fiscal appraisal is the administrative value used for the property-tax base rather than a transaction price. It changes through mass revaluations, indexation, and updates to the registered

¹Appendix E examines this spatial interpretation in two ways. Sectoral SII results and input–output measures of rural linkages show no disproportionate concentration of the aggregate association in sectors more structurally linked to rural activity. Separately, historical headquarters of a selected subset of enterprises can be linked with high confidence to the fixed urban footprints; the association remains negative within this urban-headquarters subset.

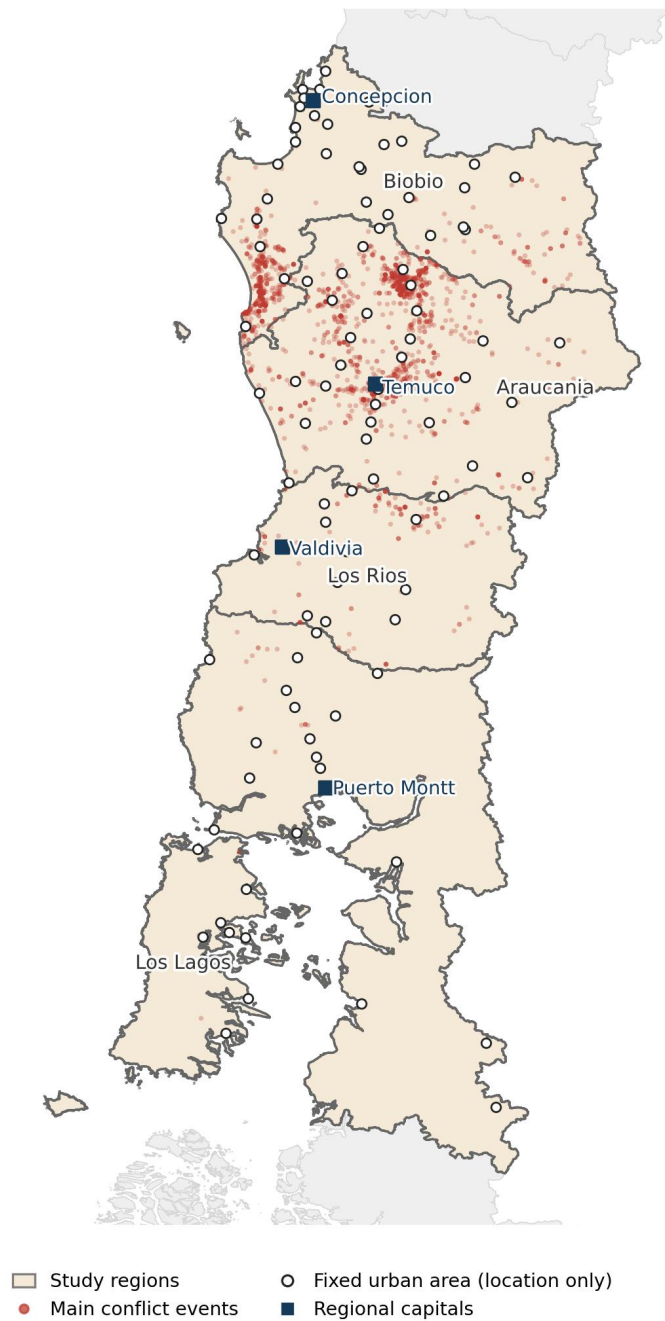


Figure 1: Main conflict events and fixed urban areas in southern Chile. The map shows attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos during 1990–2021. Point markers locate the 104 receiving fixed urban areas, with squares for the four regional capitals; marker size does not represent footprint area.

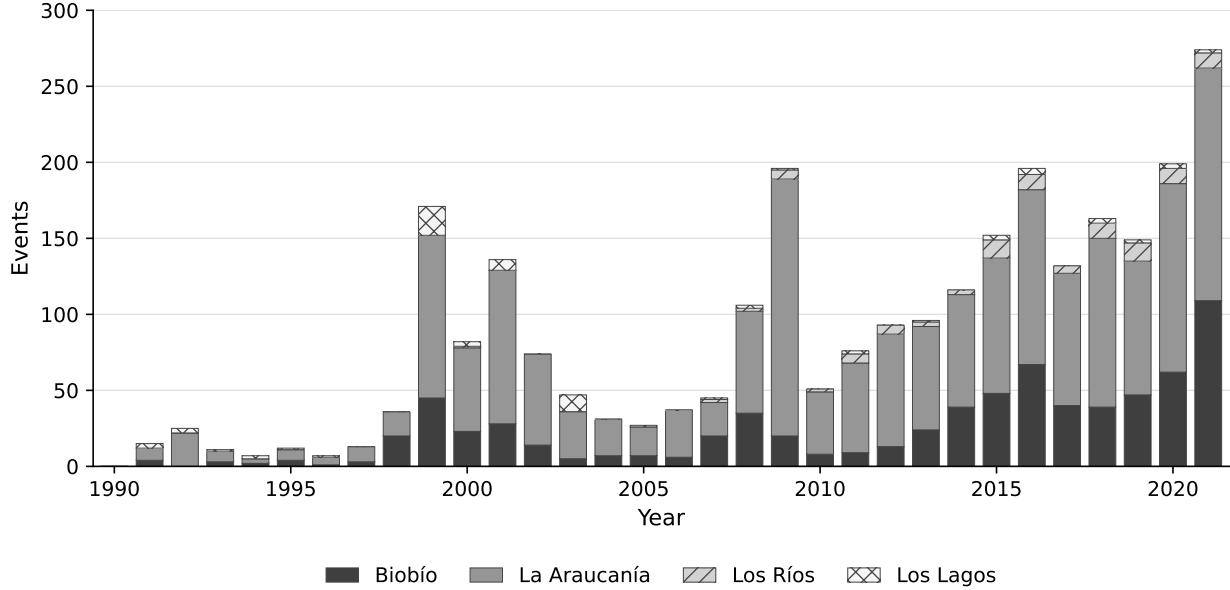


Figure 2: Annual main Mapuche conflict events by region, 1990–2021. Stacked bars count attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos.

property stock (Servicio de Impuestos Internos, 2006, 2026a,b). Fiscal appraisal measures the assessed urban property stock. Movements at longer lags may reflect administrative updates as well as changes in that stock.

WSF Evolution records changes in built surface and measures physical urban development. I use it as a complementary measure of whether conflict exposure is associated with slower physical expansion of the urban footprint; because its temporal support and interpretation differ from the headline outcomes, the detailed results are reported in the appendix.

The comparison asks whether changes in registered business activity coincide with changes measured inside the urban area. Appendix A documents each series’ construction, transformation, temporal coverage, and spatial support.

3.2 External conflict exposure

Conflict exposure comes from annual geocoded events in the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022). The event family used in the main analysis includes attacks, coercive actions, and land invasions, while protests are excluded. Let A_{mt} denote the number of eligible events occurring within 25 kilometers of urban area m in year t but outside the receiving area. External conflict exposure is defined as

$$D_{mt} = \log(1 + A_{mt}). \quad (1)$$

The baseline window uses the exact minimum straight-line distance from each eligible event point to the fixed urban-area footprint. Events outside the receiving area qualify when this distance

is at most 25 km. For these external points, distance to the footprint equals distance to its nearest boundary; calculations use the metric projection EPSG:32718. The conflict literature does not provide a standardized spatial cutoff. Closely related studies use local exposure windows in the 20–30 km range (Brancati et al., 2026; Fadare et al., 2024), while other work defines exposure along transport corridors (Alfano and Cornelissen, 2026; Liu et al., 2026). I treat the radius as an operational definition of nearby exposure rather than a structural boundary. The earlier 30 km benchmark was an intermediate point in a broader 10–60 km exploration and did not have a strong independent substantive rationale. An outcome-free road-network diagnostic indicates that short-range driving accessibility in the study geography is broadly consistent with Euclidean windows in the 20–30 km range. I use 25 km as a parsimonious geographic approximation and assess sensitivity to alternative radii from 10 to 60 km.² The benchmark is chosen independently of the outcome estimates; it does not represent an optimal radius or a literal conversion of driving minutes to kilometers.

Because exposure is defined separately for each receiving fixed urban area, the same event may contribute to several areas when their 25-km windows overlap. Among 2,775 eligible events in 1990–2021, the preferred mapping contains 8,480 event–area pairs. An event enters approximately 3.1 areas on average (median 3; maximum 9), while 109 events, or 3.9 percent, contribute to no area’s external exposure. Of these, 38 occur inside a study area and have no qualifying external receiver; their conflict counts enter separately as own-area events. Overlapping windows allow several areas to be exposed to the same event; each area-year still contributes one panel observation. Figure A3 reports the full event-level distribution in the data appendix.

The count includes events occurring outside all fixed urban areas and events occurring inside a different fixed urban area, provided that they lie within 25 kilometers of the receiving area. The measure is therefore not restricted to rural-coded events. Although the conflict studied here is predominantly rural, what defines D_{mt} empirically is that the event occurs outside the receiving urban area.

The external count can be written as $A_{mt} = R_{mt} + X_{mt}$, where R_{mt} counts events outside every fixed urban area (purely rural) and X_{mt} counts events inside a different fixed urban area. Events in R_{mt} account for 81.7 percent of receiver–event pairs. When the two components enter jointly, the outside-system coefficients are negative but imprecise for both registered enterprises and sales. The other-area sales coefficient is also negative, while its enterprise coefficient is close to zero. Appendix A reports the component estimates; the preferred exposure remains their combined count.

Main MACEDA events occurring inside the receiving area enter separately as $I_{mt} = \log(1+B_{mt})$, where B_{mt} is the annual within-area count. This allows the external-exposure coefficient to be estimated conditional on contemporaneous conflict within the area itself. Appendix A reports the event-composition, support, and spatial-assignment rules, while Appendix B examines alternative distance definitions.

²The road-network diagnostic uses a contemporary road network and imputed speeds where posted speeds are unavailable, so it is informative about spatial connectivity rather than historical travel times over the full sample period.

3.3 Empirical specification and identification

For outcome Y_{mt} , the preferred area-level specification is

$$Y_{mt} = \beta D_{mt} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} + \tau_t(X_m^0)' \theta + \varepsilon_{mt}, \quad (2)$$

where μ_m are area effects, $\lambda_{r(m),t}$ are region-by-year effects, and τ_t is linear calendar time. Standard errors are clustered by area.

The preferred specification allows secular trajectories to differ with initial population, initial rural composition, and pre-sample external conflict exposure. Population and rural share are measured for the municipality in the 2002 Census, not within the later fixed urban footprint. Population enters as $\log(1 + P_m^{2002})$. Historical conflict uses the same broad-external exact-25-km definition as contemporary exposure, summing eligible events from 1990 through 2004 and then taking $\log(1 + H_m)$. These interactions allow areas with different initial scale, rural composition, and prior conflict geography to follow different linear trajectories. Appendix B documents their construction.

Region-by-year effects absorb annual shocks shared by areas within Biobío, Araucanía, Los Ríos, and Los Lagos, including regional economic and policy conditions. The model also conditions on contemporaneous conflict inside the receiving area. The external coefficient therefore captures the association with nearby conflict conditional on direct local conflict; this co-exposure is not assumed to be exogenous.

The characteristics predate the 2005–2021 formal-activity and 2010–2021 appraisal panels. They do not predate the full lights and built-area windows, beginning in 1992 and 1990. Those longer-window models retain area and region-by-year effects and within-area conflict but omit all three trends. Post-2005 matched comparisons use all three trends for every outcome.

Unobserved local shocks could affect both conflict and economic activity. Area effects remove persistent differences, region-by-year effects remove shared regional shocks, and the trends allow trajectories to differ with initial conditions. The coefficient β uses the remaining within-area variation in exposure. It is a conditional association because this adjustment does not make subsequent conflict random.

A complementary exercise asks whether formal activity predicts subsequent conflict. One-year changes in enterprises and sales have limited predictive power for exposure one or two years ahead. With recent conflict history included, the one-year-ahead sales-level coefficient is negative but uncertain ($p = 0.075$); its interval includes zero. These projections do not resolve feedback or persistent confounding. Appendix C reports them.

Appendix D complements the area-level analysis with a within-area spatial-incidence design that compares fixed 1-kilometer grid cells within the same area-year, using grid-cell and area-by-year fixed effects. Because area-by-year fixed effects absorb aggregate area-year shocks, the exercise shows where the association is relatively more visible within an area, not whether conflict changes aggregate economic activity or the location of activity across the area.

3.4 Timing patterns

The timing analysis extends the preferred specification by including future, contemporaneous, and lagged values of the same annual exposure measure:

$$Y_{mt} = \sum_{h=1}^3 \pi_h D_{m,t+h} + \sum_{\ell=0}^{L_Y} \beta_\ell D_{m,t-\ell} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} + \tau_t (X_m^0)' \theta + \varepsilon_{mt}. \quad (3)$$

The preferred projection includes three future-exposure terms and lags through eight years for enterprises, sales, and fiscal appraisal. The nighttime-lights profile extends through ten years because of its longer temporal support.

For presentation, coefficients on future exposure are plotted at negative horizons. These terms show whether outcomes tend to move before the corresponding exposure is realized, while the contemporaneous and lagged coefficients trace how the association evolves during and after exposure. Because conflict exposure is continuous and recurrent, the exercise is best read as a description of its temporal pattern rather than as a conventional event study.

Appendix C reports additional timing evidence using spike definitions of conflict escalations as in Barrett et al. (2022) and estimates a non-absorbing difference-in-differences design (de Chaisemartin and D’Haultfoeulle, 2024).

4 Main Results

4.1 Main results

Greater external conflict exposure is associated with weaker formal business activity (Table 1). The preferred enterprise coefficient is -0.0132 (SE 0.0065; $p = 0.045$). Its 95 percent interval runs from -0.0260 to -0.0003 . The sales coefficient is -0.0320 (SE 0.0189; $p = 0.094$). Its interval runs from -0.0695 to 0.0055 . Within an area, an increase from the 25th to the 75th percentile of positive annual exposure corresponds to 2.22 percent fewer enterprises and 5.30 percent lower sales. The respective 95 percent intervals are $[-4.34, -0.06]$ and $[-11.17, 0.95]$ percent.

The progression adjusts successively for initial conditions, regional annual shocks, and direct local conflict. Adding the three predetermined trends attenuates both formal-outcome coefficients. Replacing common year effects with region-by-year effects changes enterprises from -0.0117 to -0.0134 and sales from -0.0343 to -0.0316 . Adding within-area conflict changes these estimates slightly.

The directly spatial outcomes do not show a common contemporaneous decline. Lights have a negative, imprecise full-window estimate; appraisal has positive, imprecise estimates. New built area is positive and imprecise in its longer window, reported in the appendix. Full-window lights and built-area models omit the three trends because their reference periods are not pre-sample.

Table 1: External conflict exposure and urban-area outcomes

	Formal economic activity		Urban physical outcomes		
	(1) Registered enterprises	(2) Tax-reported sales	(3) Nighttime lights	(4) Fiscal appraisal, total	(5) Fiscal appraisal, mean
Area FE + year FE	-0.020** (0.009)	-0.050** (0.022)	-0.005 (0.011)	0.001 (0.021)	0.007 (0.021)
+ core predetermined trends	-0.012** (0.006)	-0.034** (0.017)	-0.005 (0.011)	0.002 (0.016)	0.001 (0.015)
+ region-by-year FE	-0.013** (0.007)	-0.032* (0.019)	-0.004 (0.012)	0.019 (0.013)	0.016 (0.012)
+ within-area conflict	-0.013** (0.006)	-0.032* (0.019)	-0.004 (0.012)	0.019 (0.013)	0.016 (0.012)
Observations	1,768	1,768	3,060	1,248	1,248
Areas	104	104	104	104	104
Sample period	2005–2021	2005–2021	1992–2021	2010–2021	2010–2021

Notes: Coefficients are on $\log(1 + A)$, where A counts main-family broad-external events at exact point-to-footprint distance ≤ 25 km, excluding events inside the receiving area. Area-clustered SEs appear below. All rows include area effects. Row 1 uses common year effects; row 2 adds trends for 2002 municipal $\log(1 + P)$, 2002 rural share, and $\log(1 + H)$ for exact-25-km external events in 1990–2004. Row 3 replaces common year effects with region-by-year effects. Preferred row 4 adds $\log(1 + B)$ for contemporary within-area conflict. Lights use their full 1992–2021 window and omit the three trends; their rows 1 and 2 coincide. Appraisal begins in 2010 and uses all three. Outcomes enter in logs. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Post-2005 matched comparisons use the same preferred controls for all outcomes. In the common 2005–2015 built-area sample, its coefficient is positive, 0.0736 (SE 0.0350); this does not support slower physical expansion.

Appendix B reports exact radii from 10 to 60 kilometers, including 20 and 30 alongside 25 km. Only contemporary external exposure varies: the historical trend stays at 25 km. Enterprise and sales estimates remain negative throughout the range, although their precision varies.

4.2 Timing patterns

The continuous projection has negative enterprise estimates over the first several lags (Figure 3). The average at current exposure and lags 1–2 is -0.0088 (SE 0.0041; $p = 0.034$); at lags 3–5 it is -0.0108 (SE 0.0061; $p = 0.079$). The joint test of the future-exposure terms has $p = 0.590$.

Sales coefficients at current and lagged exposure are imprecise. The current and lags 1–2 average is -0.0065 (SE 0.0149; $p = 0.661$). The future-exposure test has $p = 0.115$. Absence of a clear rejection does not establish that conflict is unanticipated or exogenous.

Appraisal has negative medium-lag coefficients: lags 3–5 average -0.0220 for total and -0.0187 for mean appraisal. Future-exposure tests have $p = 0.283$ and $p = 0.393$. The administratively updated stock and observational design limit a delayed causal interpretation.

Estimated enterprise changes following major escalations are near zero and imprecise at five years under both spike definitions. Sales estimates are also imprecise. Total appraisal has a negative five-year estimate, but placebo-period movements limit interpretation. Appendix C.2 reports the

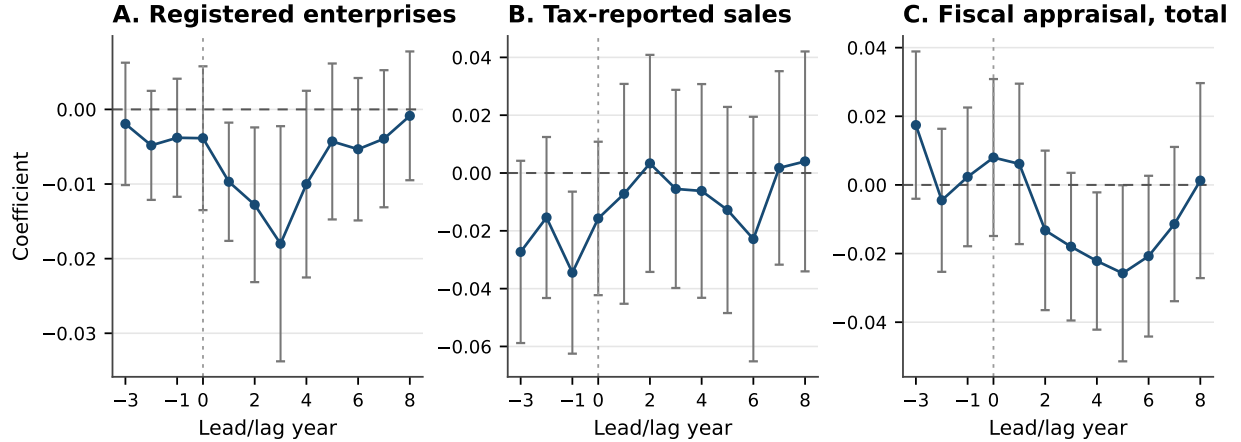


Figure 3: Timing patterns in formal activity and fiscal appraisal. The panels report coefficients from the continuous lead-lag projection defined in Section 3. Negative values on the horizontal axis are future exposure terms, zero is current exposure, and positive values are lagged exposure terms. Vertical bars are 95 percent confidence intervals based on standard errors clustered by fixed urban area. All specifications use broad external exposure within 25 kilometers, condition on contemporaneous within-area conflict, and include area and region-by-year effects and the three predetermined trends defined in Section 3.3.

definitions, comparison samples, and estimates.

4.3 Spatial interpretation and supporting evidence

Municipal tax addresses do not locate all SII activity inside the footprint. Historical headquarters of selected legal entities can be linked to parcels. The urban-headquarters coefficient is -0.0466 (SE 0.0221), while the rural-headquarters coefficient is 0.0073 (SE 0.0269). The rural estimate is uncertain; separate estimates do not establish a difference between groups. Headquarters need not identify production or sales locations. The negative urban-headquarters estimate locates the association inside the urban area for this selected group of legal entities.

Appendix D asks a different question about relative spatial incidence within the same area-year. It compares fixed 1-kilometer grid cells, holding direct local conflict constant and absorbing persistent cell differences and aggregate area-year shocks with grid-cell and area-by-year fixed effects. Selected legal entities are lower in relatively more exposed cells at the 5- and 10-kilometer radii. Commercial-use share, a measure of property-use composition, is lower at 10, 20, 25, and 30 kilometers, with a similarly negative point estimate at 5 kilometers. Nighttime lights, fiscal appraisal, and new built area do not show a common negative within-area gradient. These estimates describe relative spatial incidence, not aggregate urban decline or changes in the location of activity across the area.

The municipal association is not systematically larger in sectors with stronger predetermined agricultural and forestry input or customer linkages. This evidence does not locate SII activity inside the urban footprint, but it makes an explanation based exclusively on rural-linked sectors

less compelling. Appendix E reports the headquarters and sectoral exercises.

5 Additional Evidence

5.1 Robustness

The negative enterprise and sales coefficients persist when annual shocks are absorbed at different geographic scales. With common year effects and otherwise identical controls, the coefficients are -0.0113 (SE 0.0057) and -0.0349 (SE 0.0169). Province-by-year effects give -0.0107 (SE 0.0069) and -0.0361 (SE 0.0234). All are negative; finer geographic adjustment absorbs more treatment variation and changes precision. Appendix B also reports leave-one-province-out and exact-radius sensitivity.

The consistent asinh exercise transforms external, within-area, and historical conflict counts together. An increase from one to ten external events implies -2.13 percent for enterprises under asinh versus -2.22 under $\log(1+x)$, and -5.53 versus -5.30 percent for sales. Both transformations give similar associations for the same raw-count increase. Omitting the historical-conflict trend while retaining the other controls gives enterprise and sales coefficients of -0.0094 and -0.0347 , with intervals including zero. The enterprise estimate is therefore sensitive to adjustment for persistent prior conflict geography.

Excluding aquaculture (SII subsector 032), affected by the 2007–2010 crisis in Los Lagos (Ritter, 2025), leaves enterprises almost unchanged, from -0.0132 to -0.0132 . On the 1,066 area-years with observable subsector sales, total sales has coefficient -0.0124 , compared with -0.0144 without aquaculture. Both sales estimates are imprecise there.

Wild-cluster inference gives $p = 0.041$ for enterprises and $p = 0.079$ for sales. Spatial-HAC at 60 km gives $p = 0.013$ and $p = 0.019$, using a normal reference distribution. These procedures address sampling dependence, not confounding.

The formal-activity association is not systematically stronger where historical *Títulos de Merced* cover more surrounding land or where the 1992 Mapuche population share is higher. Appendix F reports joint estimates and multiplicity-adjusted inference.

5.2 Margins of adjustment

First registrations have coefficient -0.0144 (SE 0.0073; $p = 0.053$), in line with the lower enterprise stock. This points to fewer first formal registrations as the more relevant adjustment margin. Recorded terminations also decline in the full window, with a coefficient of -0.0319 (SE 0.0160; $p = 0.050$), rather than rising as an exit response would imply. The termination series has an administrative break in 2016–2017, and its pre-2016 estimate is imprecise. Appendix B reports these comparisons.

6 Discussion and Conclusion

This paper studies whether persistent territorial conflict is associated with weaker business activity in nearby urban areas, even when events occur outside the receiving area. In southern Chile, an increase from the 25th to the 75th percentile of positive external exposure corresponds to 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales. The enterprise interval narrowly excludes zero, while the sales interval includes a modest increase as well as a larger decline. These associations account for regional annual shocks, three pre-sample characteristics, and contemporaneous conflict inside the receiving area.

The results do not show a common contemporaneous decline in urban lights, fiscal appraisal, and newly built area. First registrations decline, while recorded terminations provide no evidence of greater exit. Negative appraisal coefficients appear at medium lags in the continuous projection, although placebo movements limit the interpretation of the escalation estimates. The formal-activity association is also not systematically stronger near historically recognized claims or in areas with a higher 1992 Mapuche population share.

Firms could respond to nearby conflict through changes in security, access to customers and suppliers, or expectations about land use and territorial disputes. The data do not distinguish these channels. The negative association among selected legal entities with urban headquarters provides evidence inside the urban area, but headquarters need not locate production or sales. The imprecise rural-headquarters estimate does not establish a difference between the two groups. Sectoral results likewise do not show a stronger association in activities with greater rural production linkages.

Regional annual effects and pre-sample conflict trends leave feedback and unobserved confounding unresolved. Municipal tax addresses also limit how precisely SII activity can be located, and the longer physical series require different trend controls because their windows begin before the baseline characteristics. Within these limits, the paper documents an economic reach beyond conflict locations that is more visible in formal business activity than in contemporaneous physical change. That reach can coexist with functioning property, tax, land-registration, and restitution institutions.

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A Data, measurement, and sample construction

A.1 Fixed urban-area geography and study sample

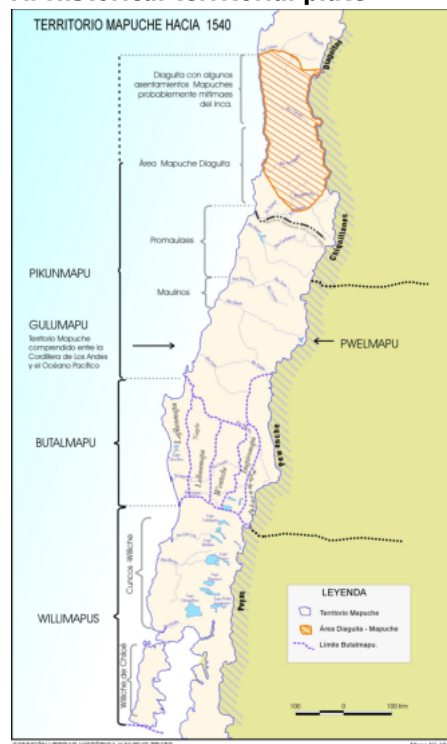
The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks (“manzanas” with AREA=1) within a municipality and is held fixed over time; its footprint can contain spatially disconnected components.

The four regions cover the part of south-central Chile where historical Mapuche and Huilliche occupation, recognized claims, contemporary restitution, and recent conflict most clearly overlap. The sample is selected for this overlap and is not nationally representative.

SII enterprise and sales records are observed at municipal tax addresses and mapped to the corresponding urban area; they therefore measure formal activity associated with the area rather than activity mechanically located inside its urban footprint.

Figure A1 provides historical and demographic context for the study corridor.

A. Historical territorial plate



B. Contemporary corridor and 1992 Mapuche shares

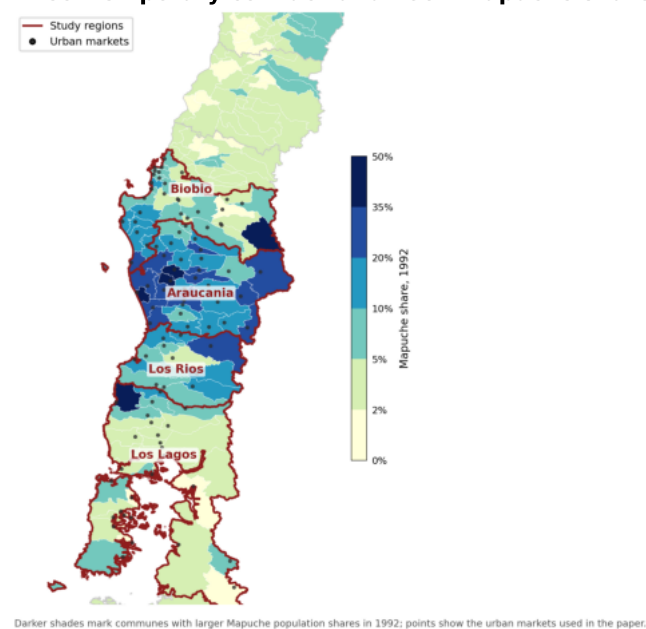


Figure A1: Historical Mapuche territory and the contemporary study corridor. Panel A reproduces a historical map of Mapuche territorial organization around 1540 from the Comisión Verdad Histórica y Nuevo Trato. Panel B shows the four-region study corridor and Mapuche population shares in 1992. Together, the panels illustrate the historical and demographic geography underlying the study area.

The business panel covers 2005–2021. Fiscal appraisal begins in 2010, while harmonized night-time lights and WSF built-area data have longer, but distinct, temporal support. Table A2 records the construction and transformation of each outcome used in the analysis.

A.2 Conflict data and event composition

Conflict events come from MACEDA. The main analysis counts attacks, state coercive actions, and land invasions; protests are excluded. This family contains 2,775 events, of which 82.1 percent have MACEDA's rural classification and 82.5 percent lie outside all fixed urban areas used to construct

the exposure measure. Most events are therefore rural, but an event qualifies as external exposure when it occurs outside the receiving urban area.

Table A1: MACEDA event composition and regional concentration

Event family	Events	Rural MACEDA	Share	
<i>Panel A. Event composition</i>				
All MACEDA events	3,730	2,587	69.4%	
Main event family: attacks, coercion, land invasions	2,775	2,278	82.1%	
Protests	955	309	32.4%	
<i>Panel B. Operational fixed-system geography of the main event family</i>				
Classification	Events	Share		
Outside all fixed urban systems	2,288	82.5%		
Inside a fixed urban system	487	17.5%		
<i>Panel C. Regional concentration of main events</i>				
Region	Main events	Event share	2017 population	Population share
Biobio	752	27.1%	1,556,805	41.8%
La Araucanía	1,847	66.6%	957,224	25.7%
Los Lagos	76	2.7%	828,708	22.2%
Los Ríos	100	3.6%	384,837	10.3%
<i>Panel D. La Araucanía share of cumulative main events</i>				
Window	Main events	La Araucanía events	Share	
1990–1992	40	30	75.0%	
1990–2002	589	401	68.1%	
1990–2017	1,990	1,371	68.9%	
1990–2021	2,775	1,847	66.6%	

Notes: The main event family comprises attacks, coercive actions, and land invasions. Rural is MACEDA’s event classification. The outside-area classification uses the fixed urban geography. Protests are not used in the exposure measure.

A.3 External exposure construction

For area m and year t , external exposure is $D_{mt} = \log(1 + A_{mt})$, where A_{mt} counts eligible events within 25 kilometers of the receiving area and outside that area. It includes events outside all fixed urban areas and events inside another fixed urban area. Events inside the receiving area enter separately as annual within-area conflict, with no radius cutoff.

A.4 Outcome construction and SII measurement

Nighttime lights, fiscal appraisal, and WSF are extracted from fixed urban footprints. Fiscal appraisal is an assessed administrative stock, not a transaction price, and is updated more slowly than a high-frequency market price. WSF records newly built area and therefore measures a physical flow. For registered enterprises and tax-reported sales, the SII address is a tax or headquarters address; it does not identify where a firm produces or where its sales take place. Appendix E examines the implication of this distinction.

A.5 External exposure components

The external count combines events outside every fixed urban system (R_{mt}) and events inside a different fixed urban area (X_{mt}). Outside-system events account for 81.7 percent of pairs. Table A3

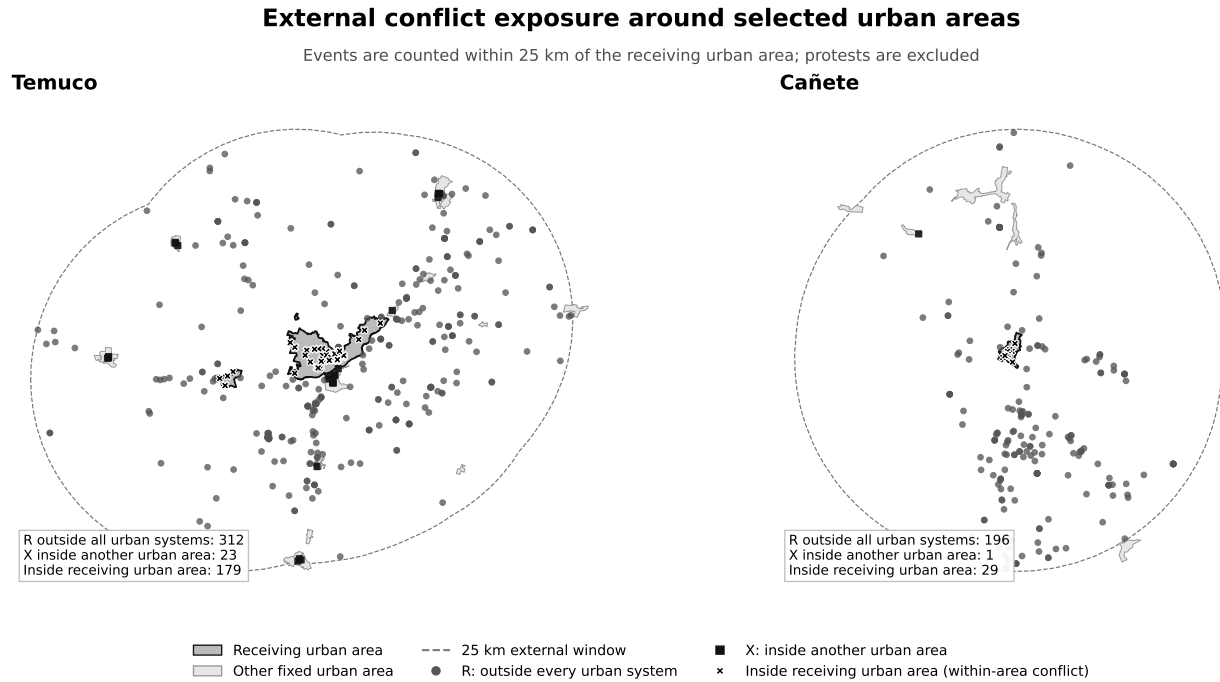


Figure A2: External conflict exposure around selected urban areas. The dashed boundary marks the 25-kilometer window around each receiving area. R events occur outside every fixed urban system; X events occur inside a different fixed urban area. Events inside the receiving area are shown separately as within-area conflict.

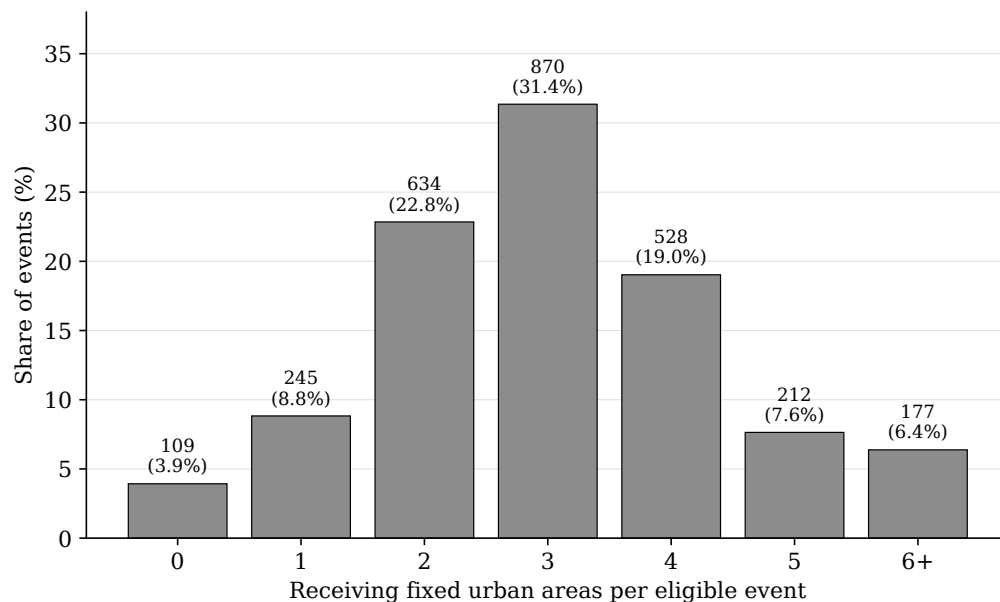


Figure A3: Event multiplicity under exact 25 km external exposure. Bars show the share of 2,775 eligible MACEDA events entering the exposure of 0, 1, 2, 3, 4, 5, or at least 6 receiving fixed urban areas. Labels report counts and shares. Own-area events are excluded from that area's external count, but may enter another area's exposure.

Table A2: Outcome construction and interpretation

Outcome	Source and spatial unit	Years	Transformation	Interpretation
Registered enterprises	SII municipality tax-address measure mapped to area	2005–2021	$\log(\text{enterprises})$	Formal enterprise stock associated with the area
Tax-reported sales	SII municipality tax-address measure mapped to area	2005–2021	$\log(\text{sales})$	Scale of formal activity associated with the area
Nighttime lights	Harmonized DMSP–VIIRS, fixed urban footprint	1992–2021	$\log(\text{total DN})$	Urban luminosity
Fiscal appraisal total	SII cadastre, fixed urban footprint	2010–2021	$\log(\text{appraised value})$	Assessed urban property stock
Fiscal appraisal mean	SII cadastre, fixed urban footprint	2010–2021	$\log(\text{mean appraisal})$	Mean assessed property value
WSF new built area	WSF Evolution, fixed urban footprint	1990–2015	$\text{asinh}(\text{new hectares})$	New built-area flow

Notes: Registered enterprises and tax-reported sales are SII municipality tax-address measures mapped to urban areas. Nighttime lights, fiscal appraisal, and WSF are constructed within fixed urban footprints. Transformations follow each outcome’s support and unit; they are not all logs.

enters $\log(1 + R)$ and $\log(1 + X)$ jointly with the preferred controls. Enterprise coefficients are -0.0139 for R and -0.0029 for X ; sales coefficients are -0.0162 and -0.0498 . Equality tests have $p = 0.242$ and $p = 0.275$. These estimates do not establish a difference between components. Preferred exposure remains $\log(1 + R + X)$.

Table A3: Components of external conflict exposure

Component	Pairs	Share of pairs	Positive area-years		
Outside-system R	6,931	81.7%	28.2%		
Other-area X	1,549	18.3%	14.6%		
Combined external A	8,480	100.0%	33.6%		
Outcome	β_R (SE)	p_R	β_X (SE)	p_X	$p(\beta_R = \beta_X)$
Registered enterprises	-0.0139* (0.0074)	0.062	-0.0029 (0.0062)	0.641	0.242
Tax-reported sales	-0.0162 (0.0218)	0.459	-0.0498** (0.0225)	0.029	0.275
Corr[$\log(1 + R)$, $\log(1 + X)$] = 0.449					

Notes: R denotes eligible events outside every fixed urban system within 25 kilometers of the receiving area; X denotes eligible events inside a different fixed urban area within the same window. Panel B enters $\log(1 + R_{mt})$ and $\log(1 + X_{mt})$ jointly in the preferred SII specification. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.6 Exposure support

Table A4 describes the support for external and within-area exposure in the full 1990–2021 area panel. Within-area conflict captures a distinct contemporaneous geography from the external-exposure measure.

Table A4: Support for external and within-area conflict exposure

Exposure measure	Mean	SD	Positive area-years
External exposure, D_{mt}	0.556	0.947	33.6%
Within-area exposure, I_{mt}	0.063	0.283	5.9%
Corr(D_{mt}, I_{mt}) = 0.374; both positive: 4.7%			

Notes: $D_{mt} = \log(1 + A_{mt})$ is broad external exposure; I_{mt} is the log of one plus annual main conflict events inside the receiving area. The panel contains 104 areas from 1990 to 2021.

A.7 Matched formal and physical samples

Table A5 re-estimates enterprises, sales, and each physical series on identical post-2005 area-years, using area and region-by-year effects, all three trends, and within-area conflict. Sales estimates are negative in each sample but uncertain. Enterprise estimates are negative, with wider uncertainty in the appraisal window. Physical estimates do not show a common negative pattern. The matched 2005–2015 built-area coefficient is positive, 0.0736 (SE 0.0350). Longer pre-2005 lights and built-area models omit the trends because the characteristics are not predetermined over those windows.

Table A5: Formal activity and physical outcomes on matched samples

Physical series	Physical coefficient	Enterprises	Sales	Common sample
Urban nighttime lights	0.0128 (0.0106)	-0.0129** (0.0064)	-0.0318* (0.0189)	1,761; 104; 17
Fiscal appraisal total	0.0187 (0.0128)	-0.0052 (0.0046)	-0.0280 (0.0181)	1,248; 104; 12
Fiscal appraisal mean	0.0156 (0.0116)	-0.0052 (0.0046)	-0.0280 (0.0181)	1,248; 104; 12
WSF new built area	0.0736** (0.0350)	-0.0180** (0.0083)	-0.0267 (0.0187)	1,144; 104; 11

Notes: Each physical outcome and both SII formal outcomes are estimated on the same rows for the indicated physical series. Coefficients are on external conflict exposure in the respective preferred specifications; standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Robustness and margins of adjustment

B.1 Trend specification and sensitivity

The preferred model contains only three differential trends, documented in Table A6. Table A7 omits only the historical-conflict trend while retaining demographic trends, area and region-by-year effects, and within-area conflict. Enterprises change from -0.0132 to -0.0094 , and sales from -0.0320 to -0.0347 . Both intervals include zero when the historical trend is omitted. Holding the remaining controls fixed shows sensitivity to adjustment for persistent prior conflict geography.

Table A6: Baseline trends in the preferred formal-activity specification

Characteristic	Reference period / source	Spatial construction	Rationale
Municipal population, $\log(1 + P_m)$	2002 INE Census	Whole-municipality population; common to its fixed urban components	Initial population scale
Municipal rural population share	2002 INE Census	Rural residents divided by all municipality residents	Initial rural composition
Historical external conflict, $\log(1 + H_m)$	MACEDA, 1990–2004	Cumulative main-family events outside receiving area and within exact 25 km	Persistent prior conflict geography

Notes: Each characteristic is standardized across 104 areas and interacted with linear calendar time. Population is $\log(1 + P)$ from the 2002 municipal census; rural share uses the same census. History sums broad-external main events in 1990–2004 at exact 25 km, then applies $\log(1 + H)$. No outcome-period events or other baseline trends enter.

Table A7: Sensitivity to omitting the pre-2005 conflict trend

Outcome	Specification	Coef.	SE	p	95% CI
Registered enterprises	Preferred	-0.0132	0.0065	0.045	[-0.0260, -0.0003]
Registered enterprises	Without conflict-history trend	-0.0094	0.0057	0.103	[-0.0208, 0.0019]
Tax-reported sales	Preferred	-0.0320	0.0189	0.094	[-0.0695, 0.0055]
Tax-reported sales	Without conflict-history trend	-0.0347	0.0189	0.069	[-0.0722, 0.0028]

Notes: Both models use the same 1,768 area-years, exact 25 km $\log(1 + A)$ exposure, area and region-by-year effects, demographic trends, and $\log(1 + B)$ within-area conflict. Only the history trend differs. SEs and 95 percent intervals use area-clustered inference.

Table A8 replaces $\log(1 + x)$ with asinh for contemporary external and within-area counts and the cumulative pre-2005 count before its trend interaction. Sample and other controls are fixed. A common increase from one to ten events gives similar enterprise and sales associations; raw coefficients are not directly comparable.

Table A8: Comparable raw-count contrasts under consistent conflict transformations

Outcome	Transformation	1 to 10 events (%)	95% CI	p
Registered enterprises	$\operatorname{Log}(1 + x)$	-2.22	[-4.34, -0.06]	0.045
Registered enterprises	Consistent asinh	-2.13	[-4.12, -0.10]	0.040
Tax-reported sales	$\operatorname{Log}(1 + x)$	-5.30	[-11.17, 0.95]	0.094
Tax-reported sales	Consistent asinh	-5.53	[-11.19, 0.50]	0.071

Notes: Percent associations equal $100\{\exp[\hat{\beta}(g(10) - g(1))] - 1\}$, with g equal to $\log(1 + x)$ or asinh . Outcomes are in logs. All three conflict quantities use the stated transformation. Both models use 1,768 area-years and area-clustered intervals.

Wild-cluster inference gives $p = 0.041$ for enterprises and $p = 0.079$ for sales (Table A9). At the 60 km spatial cutoff and three annual lags, spatial-HAC gives $p = 0.013$ and $p = 0.019$. The 30 and 90 km cutoffs give similar dependence-adjusted precision. HAC uses a normal reference distribution; area-clustered headline intervals use t with 103 degrees of freedom. Neither addresses endogeneity.

Table A9: Alternative inference for headline formal outcomes

Outcome	Coef.	Clustered SE	Wild p	Spatial-HAC SE	Spatial-HAC p
Registered enterprises	-0.0132	(0.0065)	0.041	(0.0053)	0.013
Tax-reported sales	-0.0320	(0.0189)	0.079	(0.0136)	0.019

Notes: Wild-cluster p -values use 999 repetitions and area clustering. Spatial-HAC standard errors use a 60-kilometer spatial cutoff and a three-year lag cutoff; 30- and 90-kilometer cutoffs give the same qualitative conclusion. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Geographic-time fixed effects and province influence

Table A10 compares annual effects common to all areas, effects shared within regions, and effects shared within provinces. Trends and within-area conflict are fixed. Enterprise coefficients are -0.0113 , -0.0132 , and -0.0107 ; sales coefficients are -0.0349 , -0.0320 , and -0.0361 . Finer effects absorb more treatment variation; province-year intervals are wider.

Omitting each province leaves both coefficients negative. Enterprise estimates range from -0.0195 to -0.0062 and sales from -0.0424 to -0.0195 . The reference uses region-by-year effects. Precision varies with the retained sample.

Table A10: Formal activity with geographic-by-year fixed effects

Specification	Registered enterprises	Tax-reported sales
Area + year FE	-0.0113** (0.0057)	-0.0349** (0.0169)
Area + region-by-year FE	-0.0132** (0.0065)	-0.0320* (0.0189)
Area + province-by-year FE	-0.0107 (0.0069)	-0.0361 (0.0234)

Notes: Each cell reports the coefficient on log external conflict exposure within 25 kilometers. All specifications include area fixed effects, the three predetermined trends, contemporaneous within-area conflict, and area-clustered standard errors. Region-by-year and province-by-year fixed effects replace the common year fixed effects in the corresponding rows. The sample contains 104 areas and 1,768 area-years. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

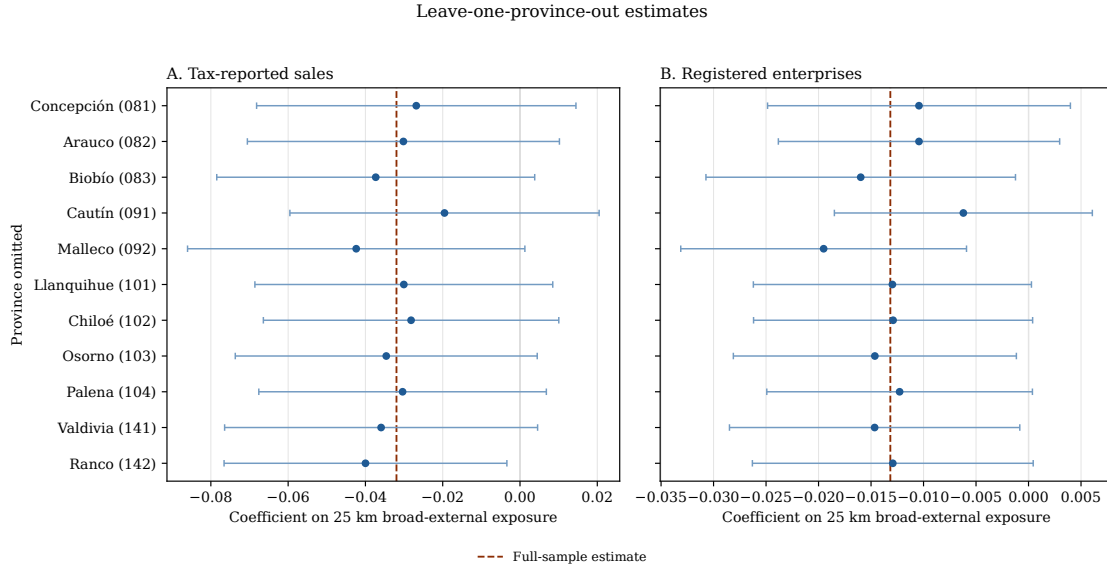


Figure A4: Leave-one-province-out estimates for formal activity. Each point reports the estimate obtained after omitting the indicated province. Vertical lines show the corresponding full-sample estimates, and horizontal bars report 95 percent confidence intervals.

B.3 Distance sensitivity

The main specification measures external conflict within 25 kilometers of each urban area. To assess whether the formal-activity results depend on this particular spatial window, Table A11 reconstructs external exposure using cumulative radii from 10 to 60 kilometers. The coefficients remain negative for both registered enterprises and tax-reported sales throughout the range. I interpret this exercise as sensitivity to the spatial window rather than as evidence for an optimal radius of influence.

Table A11: Cumulative external-conflict windows and formal activity

External window	Registered enterprises	Tax-reported sales
Within 10 km	-0.0133* (0.0078)	-0.0209 (0.0245)
Within 20 km	-0.0138** (0.0055)	-0.0292 (0.0195)
Within 25 km (preferred)	-0.0132** (0.0065)	-0.0320* (0.0189)
Within 30 km	-0.0181*** (0.0064)	-0.0478*** (0.0163)
Within 40 km	-0.0172*** (0.0062)	-0.0456** (0.0225)
Within 50 km	-0.0170** (0.0070)	-0.0414* (0.0238)
Within 60 km	-0.0166** (0.0079)	-0.0451* (0.0246)

Notes: Each row replaces the preferred 25-kilometer exposure measure with log external conflict accumulated within the indicated radius. The event family, within-area conflict, area and region-by-year effects, demographic trends, and area clustering are unchanged. History uses exact-25-km broad-external events in 1990–2004 in every row; only current external exposure varies. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Aquaculture exclusion

Los Lagos contains the largest concentration of aquaculture activity in the study corridor. Because this sector represents a large share of observed sales in the region, a sector-specific downturn could potentially contribute to the aggregate formal-activity relationship. I therefore reconstruct both SII outcomes after excluding subsector 032 (Acuicultura).

Excluding aquaculture leaves all 104 areas and enterprises almost unchanged: -0.013162 becomes -0.013152 . Suppression of subsector sales restricts the comparison to 1,066 common area-years. Total sales in that sample gives -0.0124 (SE 0.0216), compared with -0.0144 (SE 0.0216) without aquaculture. Both estimates are imprecise.

Table A12: Formal activity excluding aquaculture

Outcome	Main estimate	Excluding subsector 032	Area-years
Registered enterprises	-0.0132** (0.0065)	-0.0132** (0.0065)	1,768
Tax-reported sales	-0.0124 (0.0216)	-0.0144 (0.0216)	1,066

Notes: Aquaculture corresponds to SII subsector 032. The enterprise comparison uses the complete 104-area panel. Because subsector-level sales are suppressed in some municipality-years, the sales comparison uses the same observable sample before and after excluding aquaculture. All specifications otherwise reproduce the main area-level model. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 Margins of adjustment

The enterprise-stock result does not reveal whether formal activity adjusts through fewer entries, more exits, or both. I examine this distinction using SII records of first registrations and recorded terminations. First registrations capture new entries into the administrative registry, while terminations record the cessation of registered activity. The former is transformed as $\log(1 + x)$ because some areas record no new registrations in a given year.

First registrations give -0.0144 (SE 0.0073; $p = 0.053$); enterprise stock gives -0.0132 ($p = 0.045$). The full-window termination coefficient is also negative, -0.0319 (SE 0.0160; $p = 0.050$). It does not indicate greater exit, and administrative changes limit comparability across years.

Table A13: Enterprise stocks, first registrations, and terminations

Outcome	Transformation	Coefficient	Clustered SE
Enterprise stock	$\log(x)$	-0.0132^{**}	0.0065
First registrations	$\log(1 + x)$	-0.0144^*	0.0073
Current terminations	$\log(1 + x)$	-0.0319^{**}	0.0160

Notes: All rows use the main area-level specification and contain 1,768 area-years. The enterprise stock is measured in logs, while first registrations and terminations use $\log(1 + x)$ to retain zero-count area-years. SII termination records contain an administrative break around 2016–2017, which limits their interpretation over the full sample period. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Excluding 2016–2017 gives -0.0309 (SE 0.0164; $p = 0.063$); the pre-2016 termination estimate is -0.0116 (SE 0.0218; $p = 0.595$). Lower first registrations align with the decline in enterprise stock, while the termination estimates provide no evidence of greater recorded exit.

C Timing and identification

C.1 Continuous lead-lag results

The continuous lead-lag projection includes future, contemporaneous, and lagged values of annual external conflict exposure. For presentation, coefficients on future exposure are plotted at negative horizons. These terms capture whether current outcomes covary with exposure realized in subsequent years, conditional on the remaining exposure terms and controls. The contemporaneous and lagged coefficients describe the association with current and past exposure. Because conflict exposure is continuous and recurrent, the coefficients describe temporal patterns in the exposure–outcome relationship rather than event time around a treatment onset.

The joint future-exposure tests give $p = 0.590$ for enterprises, 0.115 for sales, 0.685 for lights, 0.283 for total appraisal, and 0.393 for mean appraisal. Newly built area shows clearer pre-movement ($p = 0.031$). Its timing profile therefore requires particular caution; absence of a rejection for another outcome does not establish exogeneity.

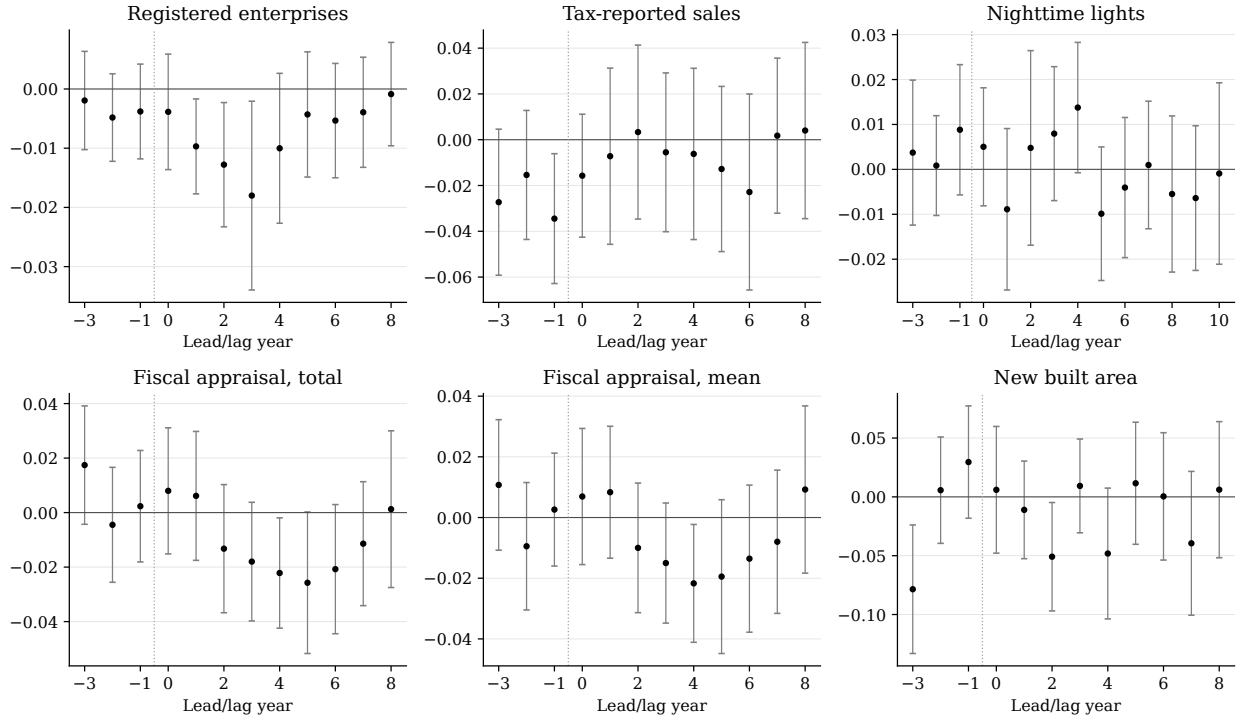


Figure A5: Continuous lead-lag profiles for urban-area outcomes. Negative horizons denote future exposure terms, zero denotes current exposure, and positive horizons denote lagged exposure. Bars show 95 percent confidence intervals based on area-clustered standard errors.

Table A14: Summary of continuous lead-lag profiles

	Sales	Enterprises	Nighttime lights	Appraisal total	Appraisal mean	New built area
Joint leads -3/-1	2.026	0.641	0.497	1.286	1.007	3.076**
	p=0.115	p=0.590	p=0.685	p=0.283	p=0.393	p=0.031
Average current/lag 0-2	-0.007	-0.009**	0.000	0.000	0.002	-0.019
	(0.015)	(0.004)	(0.006)	(0.010)	(0.009)	(0.013)
Average lags 3-5	-0.008	-0.011*	0.004	-0.022**	-0.019**	-0.009
	(0.015)	(0.006)	(0.004)	(0.009)	(0.009)	(0.014)
Average lags 6-8	-0.006	-0.003	-0.003	-0.010	-0.004	-0.011
	(0.016)	(0.004)	(0.005)	(0.011)	(0.011)	(0.015)
Observations	1,456	1,456	1,964	936	936	1,872
Areas	104	104	104	104	104	104
Window	2005-2018	2005-2018	2000-2018	2010-2018	2010-2018	1998-2015

Notes: The joint-lead row reports the Wald statistic and its p-value for the three future-exposure coefficients. Current-and-lag rows are covariance-aware averages of the indicated current and lagged coefficients. All models use area and region-by-year effects, within-area conflict, and area-clustered SEs. Formal outcomes and appraisal use all three trends; full-window lights and built-area models omit them. Windows differ with outcome availability and lead/lag support. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Major conflict escalations

The spike design isolates unusually large annual escalations in the same broad external conflict count used in the main analysis. The primary measure is an annual adaptation of the salience rule in Barrett et al. (2022). It requires a strict local peak with at least two events and an unusually high count relative to the area’s full history or recent past. Because this measure uses the following year’s count to identify a local peak and uses full-history information in two of its three thresholds, it captures isolated salience rather than a fully predetermined shock. The Past4SD comparison instead requires at least three events and a count strictly above the mean plus four sample standard deviations calculated from the preceding ten years, with all ten prior observations available.

I estimate non-normalized effects with the non-absorbing estimator of de Chaisemartin and D’Haultfœuille (2024). Event time starts at the first zero-to-one change; the observed path is retained afterwards. Adjusted models include the three preferred trend regressors and current within-area conflict. Comparisons are restricted to the same region using the estimator’s nonparametric regional adjustment rather than TWFE dummy regressors. Characteristics predate these post-2005 panels. Identification still requires conditional parallel trends and no anticipation for the supported switchers and comparison areas.

Enterprise estimates following major escalations are near zero at five years (Figure A6). Five-year enterprise estimates are 0.0061 (SE 0.0233) under Barrett and -0.0025 (SE 0.0258) under Past4SD. Sales and lights are imprecise. Total appraisal gives -0.2080 (SE 0.0724) under Barrett and -0.1631 (SE 0.0918) under Past4SD. Joint placebo tests have $p = 0.025$ and $p = 0.087$, limiting a delayed causal interpretation.

Support declines with the horizon but remains broader for registered enterprises than for appraisal. At five years, 30 Barrett and 29 Past4SD enterprise switchers contribute; none of their supported paths contains another spike by that horizon. The appraisal window begins in 2010, leaving 10 Barrett and 11 Past4SD switchers in the five-year same-switcher comparison. The smaller appraisal support and placebo-period concerns make that evidence suggestive only.

Recent economic declines do not systematically precede spike entry in Table A16. Prior conflict history has some predictive power. This describes selection into escalation and does not establish exogeneity.

C.3 Reverse-predictive evidence

Recent changes in formal activity have imprecise coefficients in projections of subsequent conflict exposure (Table A17), including the history-conditioned sales change ($p = 0.199$). The one-year-ahead history-conditioned sales level is negative with $p = 0.075$ and an interval including zero. Enterprise predictors are less precise. Feedback and confounding remain unresolved.

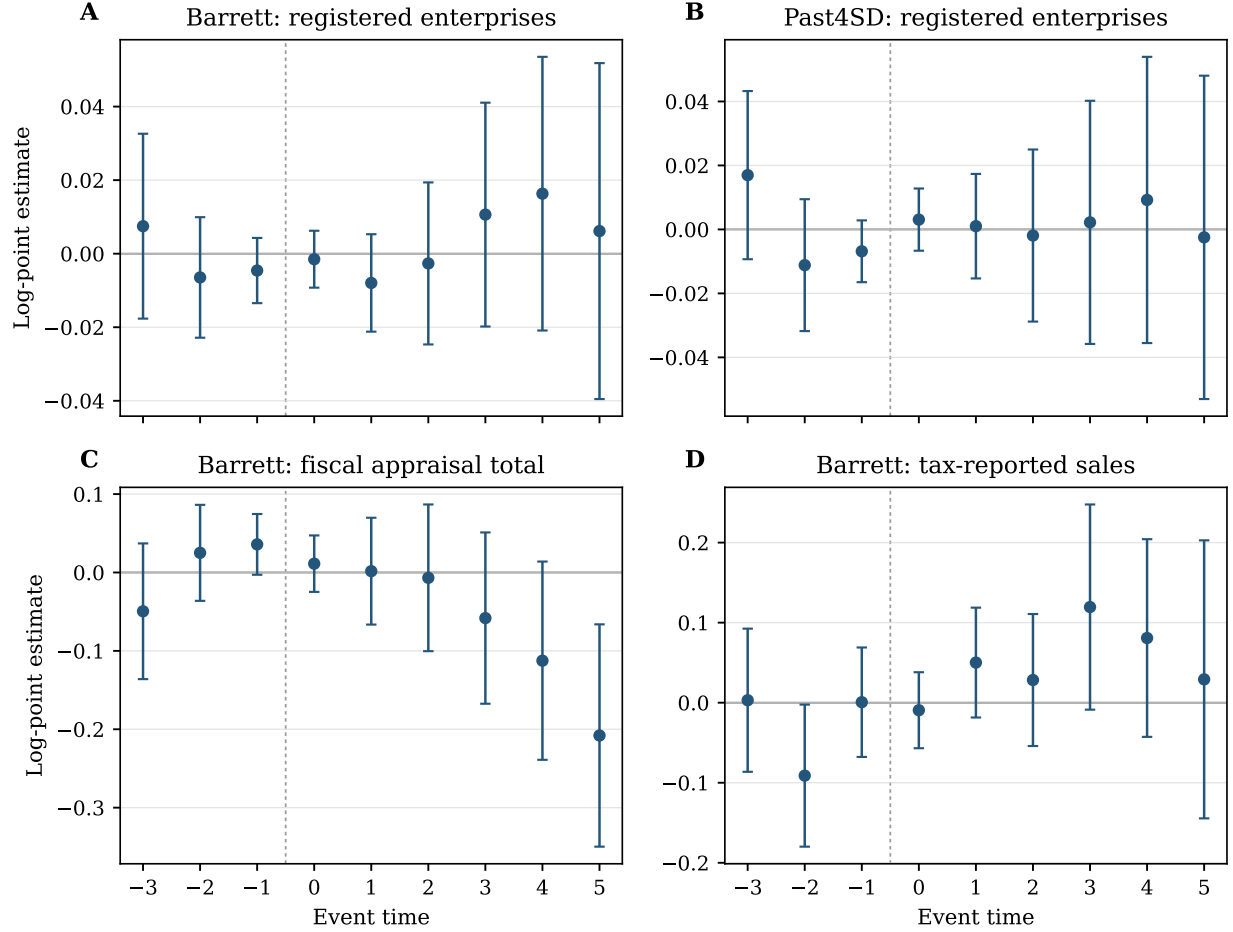


Figure A6: Major conflict escalations and urban-area outcomes. Points report non-normalized effects from the de Chaisemartin–D’Haultfoeuille estimator; vertical bars are 95 percent confidence intervals. The annual treatment remains non-absorbing after the first within-panel switch. Estimates at negative event times are placebo estimators, and the dotted vertical line separates them from post-switch effects. All specifications include the three predetermined trends and within-area conflict adjustment, with same-region comparisons as described in the text. Inference uses the estimator’s area-level variance.

Table A15: Dynamic effects following major conflict escalations

Outcome	Definition	t	+1	+2	+3	+5	Placebo p
Registered enterprises	Barrett	-0.002 (0.004)	-0.008 (0.007)	-0.003 (0.011)	0.011 (0.016)	0.006 (0.023)	0.604
	Past4SD	0.003 (0.005)	0.001 (0.008)	-0.002 (0.014)	0.002 (0.019)	-0.002 (0.026)	0.181
Tax-reported sales	Barrett	-0.009 (0.024)	0.050 (0.035)	0.028 (0.042)	0.119* (0.065)	0.029 (0.089)	0.177
	Past4SD	-0.010 (0.026)	0.037 (0.037)	-0.045 (0.044)	0.082 (0.073)	-0.022 (0.091)	0.151
Urban nighttime lights	Barrett	0.024 (0.017)	-0.011 (0.021)	-0.020 (0.024)	-0.010 (0.032)	0.024 (0.054)	0.063
	Past4SD	0.021 (0.020)	-0.015 (0.027)	-0.025 (0.032)	-0.015 (0.043)	0.036 (0.059)	0.135
Fiscal appraisal total	Barrett	0.011 (0.018)	0.002 (0.035)	-0.007 (0.048)	-0.058 (0.056)	-0.208*** (0.072)	0.025
	Past4SD	0.013 (0.025)	0.003 (0.035)	-0.020 (0.041)	-0.058 (0.084)	-0.163* (0.092)	0.087
Fiscal appraisal mean	Barrett	0.024 (0.019)	0.018 (0.035)	0.008 (0.047)	-0.040 (0.053)	-0.099 (0.081)	0.051
	Past4SD	0.017 (0.026)	0.017 (0.038)	-0.002 (0.046)	-0.018 (0.088)	-0.092 (0.095)	0.387

Notes: Cells report non-normalized DCDH estimates, with standard errors in parentheses. The final column reports the joint p -value for the three placebo estimators. Barrett is the annual local-salience composite; Past4SD uses only the preceding ten years and requires at least three events. The annual treatment path remains non-absorbing after the first switch. Adjusted models use the three preferred trends, within-area conflict, same-region comparisons, and area-level inference. Areas must have zero treatment at the initial outcome year; one area with a Barrett spike in 2005 is excluded from the non-fiscal DCDH comparison. Barrett total-appraisal, mean-appraisal, and lights placebo tests give $p = 0.025$, $p = 0.051$, and $p = 0.063$; Past4SD total appraisal gives $p = 0.087$. These concerns limit causal interpretation. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Do recent economic changes predict entry into a major conflict escalation?

	Barrett	Past4SD
<i>Panel A. One-year changes before entry</i>		
	0.033	0.073*
Enterprise growth, $t - 1$	(0.043)	(0.043)
	-0.020	-0.020
Sales growth, $t - 1$	(0.016)	(0.018)
<i>Panel B. Earlier changes</i>		
	0.054	0.008
Enterprise growth, $t - 2$	(0.057)	(0.042)
	-0.041	-0.029
Enterprise growth, $t - 3$	(0.047)	(0.038)
	0.040**	0.028*
Sales growth, $t - 2$	(0.019)	(0.014)
	-0.011	-0.001
Sales growth, $t - 3$	(0.015)	(0.013)
<i>Panel C. Predictive content of prior conflict</i>		
Year-FE R^2	0.152	0.154
Conflict-history R^2	0.176	0.166
Incremental R^2	0.024	0.012
Number of spike entries	70	45

Notes: The dependent variable is entry into a spike from an at-risk non-spike year. Economic predictors are dated strictly before spike entry. Panel A includes the two one-year changes jointly, while Panel B enters each earlier change separately. The economic-change regressions include area and region-by-year effects, the three predetermined trends, and controls for recent conflict history. Standard errors, reported in parentheses, are clustered by fixed urban area. The exercise tests whether recent economic deterioration systematically precedes escalation; it does not establish exogeneity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Do formal-activity levels or recent changes predict future external conflict?

Specification	Predictor	Horizon	Coef.	SE	p-value	N
Level	Enterprises level	t+1	-0.116	0.100	0.248	1,560
Level	SII sales level	t+1	-0.069*	0.039	0.079	1,560
Level	Enterprises level	t+2	-0.073	0.083	0.382	1,456
Level	SII sales level	t+2	-0.033	0.035	0.340	1,456
One-year change	Enterprises change	t+1	0.034	0.106	0.750	1,560
One-year change	SII sales change	t+1	-0.037	0.034	0.278	1,560
One-year change	Enterprises change	t+2	-0.030	0.103	0.768	1,456
One-year change	SII sales change	t+2	0.034	0.028	0.227	1,456
Level + conflict history	Enterprises level	t+1	-0.091	0.080	0.256	1,560
Level + conflict history	SII sales level	t+1	-0.059*	0.033	0.075	1,560
Level + conflict history	Enterprises level	t+2	-0.066	0.079	0.404	1,456
Level + conflict history	SII sales level	t+2	-0.031	0.034	0.351	1,456
Change + conflict history	Enterprises change	t+1	0.041	0.104	0.693	1,560
Change + conflict history	SII sales change	t+1	-0.047	0.036	0.199	1,560
Change + conflict history	Enterprises change	t+2	-0.035	0.103	0.733	1,456
Change + conflict history	SII sales change	t+2	0.032	0.028	0.252	1,456

Notes: Each cell is a coefficient from a projection of future external conflict exposure on the indicated enterprise or sales measure. The history-conditioned columns additionally control for current exposure and its first two annual lags. All models use area and region-by-year effects, the three trends, within-area conflict, and area-clustered SEs. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Within-area spatial incidence

The aggregate estimates describe how formal activity varies across urban areas over time. Table A18 instead compares fixed 1-kilometer grid cells within the same area-year. At a given radius R , an external event at distance d from a grid-cell centroid contributes $\max\{0, 1 - d/R\}$ to S_R . Thus, an event at the cell receives weight one, an event halfway to the stated radius receives weight one-half, and an event at or beyond the radius receives zero. S_R sums these weighted attacks, coercive actions, and land invasions outside all urban systems, and the regression uses $\log(1 + S_R)$. Each specification also includes direct local main conflict, $\log(1 + L)$, inside the same grid cell.

Grid-cell fixed effects absorb persistent differences across locations, and area-by-year fixed effects absorb shocks shared within an urban area. The selected legal-entity count is more negative at 5 and 10 kilometers, but is close to zero or positive at 20, 25, and 30 kilometers. Commercial-use share is negative at 10, 20, 25, and 30 kilometers, with a similarly negative point estimate at 5 kilometers. Nighttime lights, fiscal appraisal, and new built area do not show a common negative spatial gradient. This local within-area spatial-incidence profile does not identify aggregate urban decline, changes in the location of activity, or a causal radius. Legal-entity locations come from selected headquarters records and cannot be read as all firms or as locations of production or sales.

Table A18: Within-area spatial incidence across external radii

	Selected legal entities	Urban nighttime lights	Fiscal appraisal total	Fiscal appraisal mean	Commerce-use share	New built area
5 km	-0.1141*** (0.0275)	0.0356 (0.0475)	0.0497 (0.0902)	0.0217 (0.0637)	-0.1012* (0.0561)	-0.0010 (0.0282)
10 km	-0.0564** (0.0272)	-0.0057 (0.0582)	0.0539 (0.0704)	0.0190 (0.0594)	-0.1578*** (0.0522)	0.0467 (0.0322)
20 km	0.0419 (0.0465)	-0.0878 (0.0836)	0.1051 (0.0979)	0.0112 (0.0508)	-0.1677** (0.0717)	0.0154 (0.0574)
25 km (preferred)	0.0428 (0.0648)	-0.0659 (0.0805)	0.1332 (0.0920)	0.0305 (0.0526)	-0.1618** (0.0629)	0.0161 (0.0440)
30 km	0.0446 (0.0783)	-0.0452 (0.0779)	0.1536* (0.0851)	0.0606 (0.0507)	-0.1637*** (0.0614)	0.0230 (0.0364)
Observations	27,591	27,591	19,014	19,014	15,901	17,853
Fixed 1-km grid cells	1,623	1,623	1,623	1,623	1,623	1,623
Areas	104	104	104	104	104	104
Outcome years	2005–2021	2005–2021	2010–2021	2010–2021	2012–2021	2005–2015

Notes: Each column includes fixed effects for 1-kilometer grid cells and area-by-year fixed effects. For radius R , the R-only external exposure assigns an eligible event at distance d a weight $\max\{0, 1 - d/R\}$: one at the grid cell, one-half halfway to R , and zero at or beyond R . $\log(1 + S_R)$ sums these weighted attacks, coercive actions, and land invasions outside all urban systems. Direct local main conflict, $\log(1 + L)$, inside the same grid cell is included in every specification. Standard errors are clustered by fixed urban area. Coefficients and standard errors are reported to four decimal places. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Municipal-versus-urban interpretation

E.1 Registered legal entities

The SII outcomes used in the main analysis are assigned through municipal tax addresses and therefore do not mechanically locate economic activity inside the fixed urban footprints. I first

examine this measurement issue using registered legal entities (*personas jurídicas*) whose historical headquarters can be matched with high confidence to cadastral parcels.

The urban-headquarters coefficient is -0.0466 (SE 0.0221); rural headquarters give 0.0073 (SE 0.0269). Zero-count area-years are retained. The uncertain rural result does not establish an urban-versus-rural difference or an exclusively urban association. The samples are selected, and headquarters need not identify production or sales locations.

Table A19: External conflict and registered legal entities with geocoded headquarters

Outcome	Transformation	Coefficient	Clustered SE	Zero area-years
Registered enterprises	$\log(x)$	-0.0132**	0.0065	0
Total registered legal entities	$\log(x)$	-0.0073	0.0107	0
Temporally assigned legal entities	$\log(x)$	-0.0131	0.0112	0
Cadastral-classified legal entities	$\log(1+x)$	-0.0335	0.0230	109
Cadastral urban legal entities	$\log(1+x)$	-0.0466**	0.0221	152
Cadastral rural legal entities	$\log(1+x)$	0.0073	0.0269	913

Notes: Historical headquarters are matched to cadastral parcels using the administrative address records. The linked samples are selected and headquarters locations need not identify production or sales locations. Zero-count area-years are retained. Standard errors are clustered by area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.2 Rural production linkages

A second concern is that municipality-level SII outcomes could primarily reflect businesses tied to the surrounding rural economy rather than activity associated with urban areas. I examine this possibility using predetermined sectoral linkages from the Chilean input-output matrices (*Matriz Insumo-Producto*, MIP) published by the Banco Central de Chile.

For each sector, I construct two measures of its direct connection to rural production. The rural input linkage is the share of the sector’s domestic intermediate inputs purchased from agriculture, fruit growing, livestock, and forestry. The rural customer linkage is the share of the sector’s domestic intermediate sales sold to those same rural activities. Fishing and aquaculture are excluded from the rural group. The 2003 MIP is the preferred benchmark because it is the closest pre-period matrix to the 2005–2021 estimation window; the 1996 MIP provides an earlier benchmark. The linkage measures are standardized across the ten comparable non-rural sectors retained in the preferred sector panel.

Let L_s denote either standardized linkage for sector s . I estimate

$$Y_{mst} = \beta (D_{mt} \times L_s) + \rho (I_{mt} \times L_s) + \alpha_{ms} + \gamma_{st} + \delta_{mt} + \varepsilon_{mst}, \quad (4)$$

where D_{mt} is external conflict exposure and I_{mt} is within-area conflict. Area-by-sector fixed effects absorb persistent differences in local sector composition, sector-by-year fixed effects absorb national sectoral shocks, and area-by-year fixed effects absorb all shocks common to sectors in the same area and year. The coefficient β therefore asks whether the formal-activity association is

systematically different in sectors that are more structurally connected to rural production.

Table A20: External conflict and predetermined rural sectoral linkages

Benchmark	Linkage	Sales interaction	Enterprise interaction
2003	Customer	-0.0168 (0.0279)	0.0046 (0.0050)
2003	Input	-0.0008 (0.0382)	-0.0034 (0.0034)
1996	Customer	-0.0167 (0.0378)	0.0021 (0.0054)
1996	Input	0.0033 (0.0369)	-0.0028 (0.0036)

Notes: Input and customer linkages are constructed from the Banco Central de Chile input-output matrices and standardized across the ten sectors in the preferred sample. The 2003 matrix is the main pre-period benchmark and the 1996 matrix provides an earlier benchmark. Each coefficient is the interaction between log external conflict exposure within 25 kilometers and the indicated standardized sectoral linkage. Specifications also include the corresponding interaction with within-area conflict and absorb area-by-sector, sector-by-year, and area-by-year fixed effects. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction estimates are small and do not display a systematic negative gradient for either sales or enterprises. The conclusion is similar using the 2003 input and customer linkages and the earlier 1996 benchmark. The aggregate formal-activity association therefore does not appear to be disproportionately concentrated in sectors with stronger structural links to rural production.

The headquarters records locate selected legal entities inside the urban area, while the input-output results show no stronger association in sectors with greater rural production linkages. Neither exercise locates all SII sales or production.

F Historical heterogeneity

I examine whether the association with contemporary external conflict varies with two historical characteristics of the receiving area. Historical territorial exposure is measured by the share of terrestrial land in the external 0–25 km catchment covered by *Títulos de Merced* (TDM). Historical Mapuche presence is measured by its share of the urban population in 1992, reconstructed from individual census records and harmonized to the fixed urban-area geography. The two measures capture related but distinct dimensions of the historical setting and enter jointly in the specifications.

Table A21 reports the 25 km interactions. The TDM enterprise interaction is -0.0089 (SE 0.0110) and the sales interaction is -0.0007 (SE 0.0210). Both are uncertain, as are the 1992 Mapuche-share interactions. Figure A7 varies only the moderator catchment through 10, 20, 25, 30, and 50 km. Contemporary conflict exposure and cumulative pre-2005 external conflict remain defined at 25 km. These estimates do not establish a systematic historical gradient.

The preferred TDM total-appraisal interaction is -0.0189 (wild $p = 0.319$); mean appraisal gives -0.0087 ($p = 0.587$). Neither survives multiplicity adjustment. Appraisal Mapuche interactions are positive and uncertain. A supplementary 30 km conflict/TDM sensitivity allows conflict slopes to vary by region. Appraisal is an assessed stock rather than transaction prices. TDM coverage is

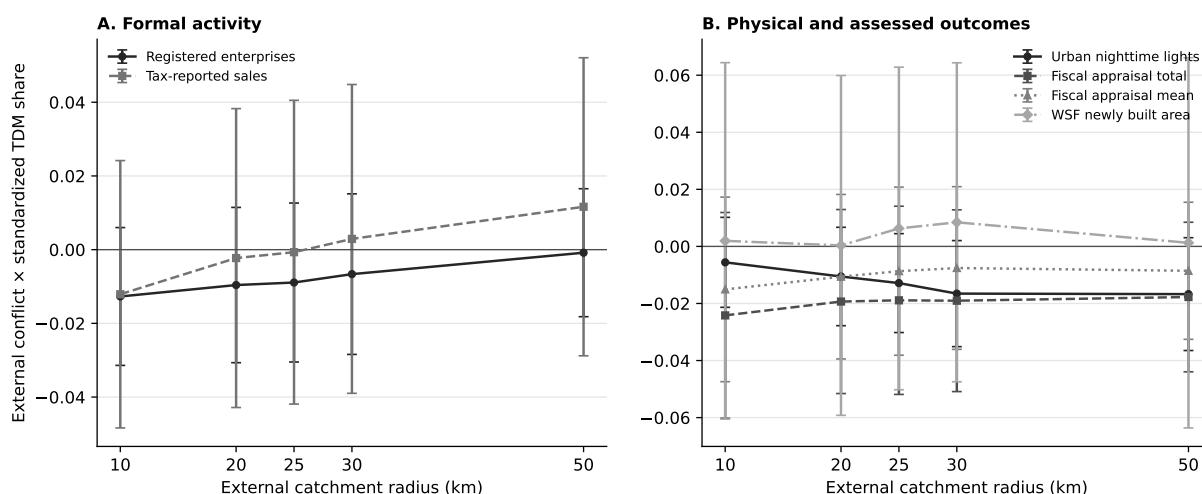
concentrated: 43 of 104 areas have no titled land in the external 25 km catchment, and much of the positive variation is in Araucanía.

Table A21: Historical territorial exposure, baseline Mapuche presence, and urban-area outcomes

Outcome	TDM share, external 0–25 km			1992 Mapuche urban-population share		
	β (SE)	Wild p	BH q	β (SE)	Wild p	BH q
<i>Panel A. Formal activity</i>						
Registered enterprises	-.009 (.011)	.512	.902	.010 (.015)	.663	.909
Tax-reported sales	-.001 (.021)	.975	.991	-.004 (.033)	.918	.975
<i>Panel B. Physical and assessed outcomes</i>						
Urban nighttime lights	-.013 (.009)	.174	.902	.007 (.015)	.682	.909
Fiscal appraisal total	-.019 (.017)	.319	.902	.036 (.024)	.223	.909
Fiscal appraisal mean	-.009 (.015)	.587	.902	.019 (.022)	.469	.909
WSF newly built area	.006 (.029)	.866	.991	-.033 (.038)	.460	.909

Notes: The table reports interactions between external conflict exposure and two standardized historical moderators, estimated jointly. TDM exposure is the share of terrestrial land in the receiving area's external 0–25 km catchment covered by historical *Títulos de Merced*. Mapuche population share is reconstructed from individual records of the 1992 Census and measures the Mapuche share of the historical urban population harmonized to the fixed urban-area geography. The joint sample contains 102 areas; Curaco de Vélez and Palena have no urban population records in the 1992 census geography. All models include area and region-by-year effects, within-area conflict, and moderator-specific trends. Post-2005 panels use the three preferred trends; the longer built-area panel omits them. Lights here use the post-2005 historical-moderator sample. Models also include the corresponding moderator interactions. Standard errors are clustered by fixed urban area and wild-cluster p -values are reported. TDM q -values adjust the 30 radius-by-outcome tests across the five stated catchment boundaries; Mapuche q -values use the pre-specified Mapuche interaction family. Stars use the reported wild-cluster p -values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A7: Historical TDM exposure across alternative catchment boundaries



Notes: Points report interactions between external conflict exposure and standardized TDM land share; bars show 95 percent confidence intervals based on area-clustered standard errors. The 10-, 20-, 25-, 30-, and 50-km catchments are alternative boundaries for the same measure of surrounding historical territorial exposure and are highly correlated. Contemporary conflict exposure uses the preferred 25-km definition in every regression; only the historical moderator catchment varies.

G Institutional response: CONADI

CONADI records provide evidence on formal land-acquisition activity in the same territorial setting. Article 20(a) and 20(b) acquisitions are aggregated to area-years using the official municipality codes and the municipality–urban-area crosswalk. The source reports an acquisition year, although the precise administrative stage represented by that date is not documented. The two outcomes are the log of one plus the number of recorded acquisitions and the log of one plus reported acquired hectares.

Greater external conflict exposure is followed by more recorded CONADI acquisition activity at the two-year lag; the one-year coefficients are imprecisely estimated. The cumulative current-plus-two-lag association is positive for both acquisitions and acquired hectares. This documents continued formal land-acquisition activity in conflict-exposed territories; it does not identify the effect of CONADI acquisitions on conflict or local economic activity.

Table A22: External conflict exposure and recorded CONADI land acquisition

Outcome	Static	D_t	D_{t-1}	D_{t-2}	Joint p	Cumulative
Acquisition events	0.0751** (0.0320)	0.0675* (0.0343)	-0.0051 (0.0295)	0.0896*** (0.0263)	0.003	0.1519*** (0.0442)
Reported hectares	0.1707* (0.0873)	0.1582* (0.0935)	-0.0525 (0.0923)	0.2456*** (0.0871)	0.022	0.3513** (0.1378)

Notes: The static model uses contemporaneous external conflict exposure. The distributed-lag model includes exposure at t , $t - 1$, and $t - 2$ jointly. The cumulative column reports the sum of these three coefficients, with its covariance-adjusted standard error. All specifications use the preferred 104-area, 2005–2021 specification and area-clustered standard errors. CONADI outcomes are transformed as $\log(1 + x)$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.